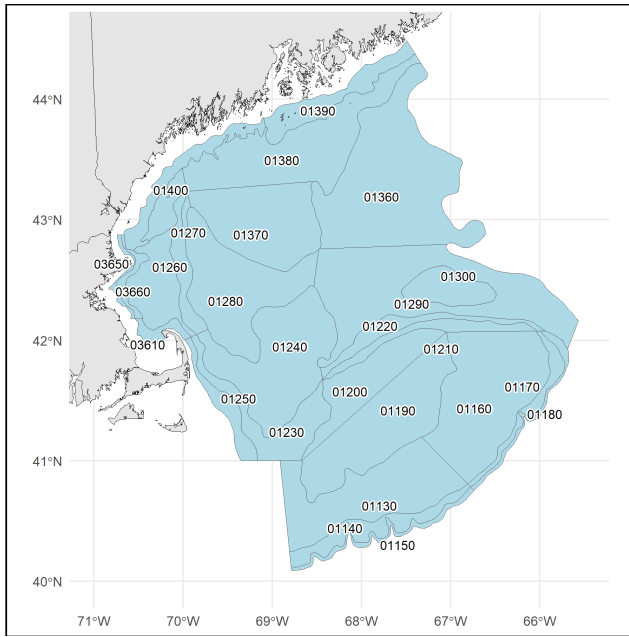


American Plaice (*Hippoglossoides platessoides*) Snapshot Ecosystem & Socioeconomic Profile

Fall 2026

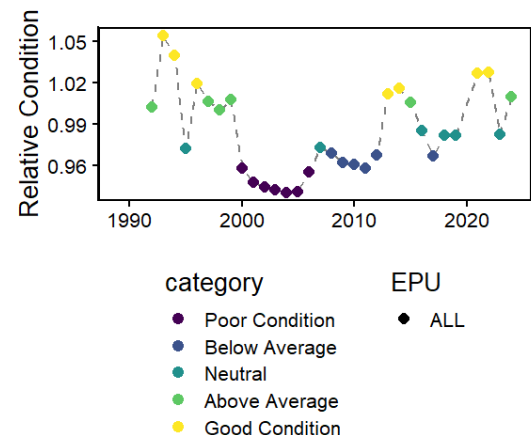


This snapshot ESP provides an update to ecosystem indicators identified in Term of Reference 1 of the 2022 American Plaice Research Track Assessment (J. Behan and Kerr 2022) (J. Behan et al. 2022). (J. Behan and Kerr 2022) identified a suite of ecosystem indicators that are relevant to American plaice. Recruitment of plaice is related to sea surface temperature, the Atlantic Multidecadal Oscillation (AMO), and the North Atlantic Oscillation (NAO). Bottom temperature anomalies (also affected by the Gulf Stream Index) and spawning stock biomass (SSB) were both significant factors influencing plaice shift to deeper and more northward distribution. The AMO was also prevalent variable in American plaice condition. The goal of this report is to provide recent context for how these indicators may impact the American plaice stock, as well as highlighting recent socioeconomic information, using time series with the most recent years of data since the 2022 Research Track Assessment.

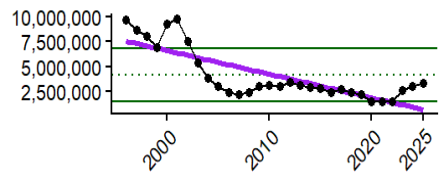
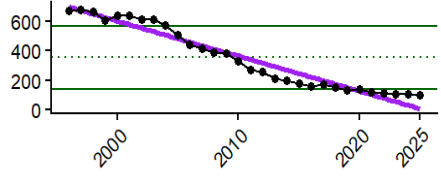
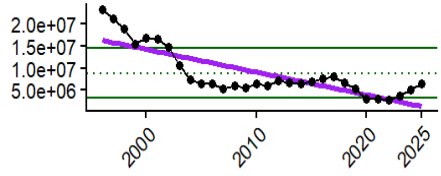
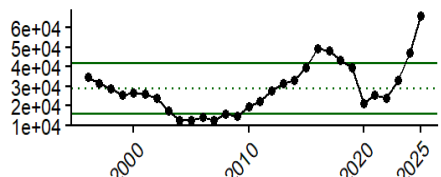
In the 2016 Climate Vulnerability Assessment, American plaice had an overall low vulnerability rank but high climate exposure from increasing surface temperatures and ocean acidification (Hare et al. 2016). There is evidence of American plaice shifting distribution into deeper waters (Nye et al. 2009) and increasing latitude in the Gulf of Maine (J. T. Behan and Kerr 2026), and further warming may decrease available habitat.

The New England Fishery Management Council (NEFMC) put American plaice under a rebuilding plan in 2004 (New England Fishery Management Council 2003) after significant declines in landings and revenue. While the number of commercial vessels landing plaice has continued to decline, revenue and landings have rebounded somewhat post-COVID19 (New England Fishery Management Council 2026) along with spawning stock biomass (North-east Fisheries Science Center 2024).

Condition of American plaice across the Northeast Shelf exhibited lows in the 2000s to 2010, indicating thinner, lower-quality fish. Condition values have increased overall since 2010, with most recent years having neutral to good condition, and are experiencing fewer years of poor or below-average condition compared to lows in the 2000s.



Indicator Units	Recent Status	Implications	Time Series
Bottom temperature (annual prior to Fall and Spring surveys, °C)	Near the long term mean (fall and spring)	Since 2023, annual mean temperatures prior to the fall survey (top; October - September, bottom; March - February) have declined from recent highs and are near the long-term mean. An overall warming trend in bottom temperatures may eventually result in changes in plaice distribution and condition. Bottom water temperatures control spawning - optimum spawning temperatures range between 3-6C but can thrive from -0.5 to 13C. Spring bottom temperatures are also correlated with mean latitude of plaice distribution. This data set does not yet include 2026.	
Gulf Stream Index (annual mean prior to Fall survey)	Near the long term mean	The annual mean of the Gulf Stream position leading up to fall survey affects recruitment. When the strength of Labrador current is weak, a higher proportion of warm water coming from the Gulf Stream enters the Gulf of Maine, increasing Gulf of Maine water temperatures. There is an overall long-term increasing trend in the position of the Gulf Stream since 1970, with the Gulf Stream being further north than the long-term mean in the 2010s, which may be contributing to warmer water temperatures in the Gulf of Maine and affecting plaice recruitment.	
Sea Surface Temperature (Mar-June)	Near the long term mean	March-June SST anomaly affects plaice recruitment, as the thermal limit for survival and incubation of American plaice eggs is 14C. Spring temperatures have hovered around 7-8°C since the early 1980s, however temperatures in the early 2020s exceeded the long-term mean. Seasonal temperatures have declined since 2022, which may be beneficial for plaice recruitment. However, a long-term increasing trend in sea surface temperatures may lead to overall declines in recruitment.	
AMO (6 months prior to Fall and Spring survey)	Above the long term mean (fall and spring)	The AMO index has been above the long-term mean since 2020. AMO values below 0 and above 0.25 are associated with lower recruitment, while values between approximately 0-0.25 are associated with higher recruitment (Behan and Kerr 2022, fig. 2). Recent AMO values are positive, with some being above the transition point of ~0.25, so impacts on recruitment may be mixed.	

Indicator Units	Recent Status	Implications	Time Series
Commercial landings (millions of lbs)	Near the long term average	Commercial landings of American plaice exhibited a sharp decline in the early 2000s with a long-term decreasing trend (purple). Trends in landings follow the implementation of a rebuilding plan for the stock in 2004 and successful rebuild in 2019.	
Number of commercial vessels (# of federally-permitted vessels landing over 1lb of American plaice)	Below the long term average	The number of commercial vessels landing American plaice has declined steadily from highs in the late 1990s, with a long-term decreasing trend (purple).	
Commercial revenue (millions, inflation adjusted to 2025 USD)	Near the long term average	Commercial revenue declined sharply after 2000 with a long-term decreasing trend (purple), likely due to a similar trend in landings. Average price per pound (not shown) dropped 50% from 2016 to 2023 and may also be a driver of declining revenue.	
Commercial revenue per active American plaice vessel (2025 USD)	Above the long term average	Revenue per vessel gradually increases from lows near 2010 to highs above the long-term mean prior to 2020. A steep decline in revenue per vessel in 2020 likely stems from the COVID-19 pandemic. In recent years, revenue per vessel has risen greatly to above the long-term mean, a possible result of a continued decline in vessels but slight increase in landings.	

* The y-axis units are included in the “Indicator” column of the table. In all figures, the dashed line represents the time series mean, and the solid green lines indicate ± 1 standard deviation. Commercial data were derived from the commercial dealer database hosted at the Greater Atlantic Regional Office. All dollar values have been adjusted to 2026 real dollars using the [Gross Domestic Implicit Price Deflator](#).

We welcome your observations! Please contact northeast.ecosystem.highlights@noaa.gov with any on-the-water insights or changes observed in the American plaice fishery and nefsc.esp.leads@noaa.gov with questions or comments on the information presented in this report.

The code used to create this report can be viewed online: github.com/NEFSC/READ-EDAB-plaiceESP2026

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